

ESTIMATION OF LINEAR TREATMENT EFFECTS FROM MATCHED PAIR DATA

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Summary: In the analysis of matched pairs it is standard practice to reduce the data to pairwise differences and to identify the mean difference with the treatment effect. With such an approach only those effects that manifest in a shift of one of the marginal distributions with respect to the other can be detected efficiently. In this paper a more general measure of treatment effect, first proposed by Lehmann (1974), is considered in the case where the treatment effect consists in a change of both location and scale.

1. Introduction

A lay definition of a treatment effect is that it is the resulting change in an object of study that can be ascribed to the administration of a treatment. Although this definition gives one an intuitive idea of what is meant by the term “treatment effect”, it is by no means a precise statistical definition. In the present context, two objects which are judged to be similar based on certain background information, are randomly assigned to serve as a control or to receive a treatment. Denote by X and X' the measurement of the characteristic of interest prior to the application of any treatment. Since the objects are randomly assigned to the treatment or control group, it is reasonable to expect the pair (X, X') to have a symmetric distribution, that is, (X, X') has the same distribution as (X', X) . However, one does not observe the pair (X, X') , but rather the pair (X, Y) where $Y = q(X')$ denotes the value of $MSC2000: 62G10$

X' after the treatment has been applied. There is no treatment effect if and only if $q(x) = x$, that is, when (X, Y) has the same distribution as (Y, X) . In terms of the joint CDF H of (X, Y) ,

$$H(x, y) = H(y, x) \text{ for all } x \text{ and } y;$$

see Lehmann (1986, Section 6.8).

The description of the nature of a treatment effect is a problem of some practical importance. It is standard practice to express the treatment effect in terms of the degree of asymmetry around zero of the marginal distribution of $Y - X$; see Lehmann (1986, Section 6.8). Suppose, however, that (X, X') has a standard bivariate normal distribution and that $Y = 2X'$. Then the distribution of $Y - X$ is symmetric around zero, indicating no treatment effect while, in fact, the treatment has affected the spread of X' . Thus, a measure of treatment effect that is more widely valid than asymmetry in the distribution of $Y - X$ is desirable.

Towards establishing such a measure, notice that H can be written in the form

$$H(x, y) = C(F(x), G(y))$$

where

$$C(u, v) = H(F^{-1}(u), G^{-1}(v))$$

is the so-called copula. The copula is a bivariate distribution function concentrated on the unit square. Now, it is easy to show that H is symmetric in its arguments if and only if C is so *and* F and G are equal. In many instances, it is reasonable to assume that C is symmetric because of the randomized treatment/control allocation. A test of the hypothesis that C is symmetric can also be made – see Lombard (2005). If C is indeed symmetric,

it follows that there is no treatment effect if and only if the marginal distribution functions F and G are identical. Therefore, a function such as

$$q(x) = G^{-1}F(x), \quad (1)$$

which compares the marginal distributions of X and Y , could be an appropriate measure of treatment effect. We will refer to this function as the QMF (quantile matching function). The nomenclature comes from the fact that setting $x = F^{-1}(p)$ in equation (1) gives

$$q(F^{-1}(p)) = G^{-1}(p), \quad (2)$$

so that q matches the quantiles of the X distribution to those of the Y distribution. The QMF can be viewed as the population version of a Q-Q plot.

Lehmann (1974) proposed the QMF as a measure of treatment effect in the context of fully randomized experiments, i.e. where the X and Y samples are independent and sample sizes may be different. Doksum (1974) studied the properties of the QMF, referring to it as the horizontal shift function. The QMF was further studied by Doksum (1975), Doksum and Sievers (1976) and Switzer (1976) where procedures for testing the hypothesis of no effect were established. Henry, Wells and Tiwari (1994) considered independent censored data sets. Einmahl and McKeague (1999) extended the methodology to higher dimensions using an empirical likelihood ratio and also considered the possibility of censored data. These authors all restricted attention to independent samples. Lombard (2005) was apparently the first to consider matched pair data. He constructed fully nonparametric confidence bands for the QMF using a Kolmogorov-Smirnov type statistic. Wilcox (2006) considered matched pair data where ties among the data are possible.

The confidence bands for q produced by fully nonparametric methods are typically rather wide unless the sample sizes involved are quite large.

Furthermore, the bands typically do not cover the full span of the data, often degenerating to $-\infty$ or $+\infty$ near the extremes. One would expect narrower bands if some further assumptions regarding the underlying distributions were justified.

The main focus in the present paper will be on linear QMFs, i.e.

$$q(x) = \gamma_0 + \gamma_1 x \quad (3)$$

with $\gamma_1 > 0$. From (2) and (3), we have

$$G^{-1}(t) = \gamma_0 + \gamma_1 F^{-1}(t).$$

Hence, the statement that q is linear is equivalent to the statement that X and Y belong to the same location-scale family of distributions. In this case, the definition of no effect reduces to stating that $\gamma_0 = 0$ and $\gamma_1 = 1$. That is, the QMF is the identity function. As such, the function

$$\Delta(x) = q(x) - x = \gamma_0 + (\gamma_1 - 1)x \quad (4)$$

gives a succinct description of the discrepancy between the marginal CDFs F and G . Doksum and Sievers (1976) considered parameter estimation for a linear QMF in the special case where the marginal distributions are normal, sample sizes are equal and samples are independent. Hsieh (1996) proposed a weighted regression method for estimating the parameters in a linear QMF, again for independent samples. These estimators were also used to test the hypothesis of no effect. None of these authors considered the question of testing for linearity of the QMF.

Our objectives in this paper are twofold. Given n pairs of values $(X_1, Y_1), \dots, (X_n, Y_n)$ from a matched pair experiment, first test the linearity assumption (3) and, second, assuming linearity construct a simultaneous confidence band for Δ . The paper is structured as follows. In Section 2 the application that motivates the present work is discussed briefly. In Section

3, two tests for linearity of the q function are presented and their powers compared. Section 4 discusses the construction of a simultaneous confidence band for a linear $\Delta(x)$. The band is typically substantially narrower than those produced by the fully non-parametric methods developed by Lombard (2005). In Section 5, the proposed methodology is illustrated by application to some data.

We alert the reader to the fact that there is another shift function, called the vertical shift function and defined as $G(F^{-1}(t))$, which can be looked upon as the population version of a P-P plot. Several authors have studied its properties. Li, Tiwari and Wells (1996), for instance, used it to compare two distributions for independent samples from both uncensored and censored data. The statistical properties of estimators of the vertical shift function differ considerably from those of estimators of the horizontal shift function considered here. We do not undertake a comparison of the two types of shift function in the present paper.

In concluding this introductory discussion we remark that while most of the formal theoretical results in this paper are of an asymptotic nature, their practical implementation rests on the use of permutation and bootstrap methods. Our reference for the basic methodology is Davison and Hinkley (1997).

2. Application data

Abrasiveness is an important quality characteristic of coal. A standard method of measuring abrasiveness is known as the YGP method, after Yance, Geer and Price (1951). The determination of abrasiveness by this method is subject to considerable variability inherent in the measurement apparatus. The apparatus consists of a rotary coal grinder (more or less the size of a typical

household food processor) fitted with four stainless steel plates. The observed abrasiveness value is defined to be the loss of mass (in milligrams) from the plates after grinding a coal sample of specified mass for a preset length of time. Two subsamples were prepared from each of 98 distinct batches of coal and an abrasiveness determination was made on each subsample, one using the YGP method, the other using a modified version of it that incorporated both changes in procedural specifications and a differently constructed grinder. The modified method is expected to yield results that are less variable than those produced by the YGP method. In this example the members of a pair (X, Y) are correlated because they arise from a batch of coal that has common quality characteristics, distinct from any of the other batches. In fact, the 98 coal batches originated from two different mines. Figure 1 shows kernel estimates of the marginal densities of the X and Y values. From the figure it is already apparent that the spread of Y -values is somewhat less than that of the X -values. It is also apparent that the data are not from a normal distribution.

Denote the order statistics of the X and Y samples by $X_{(1)} < \cdots < X_{(n)}$ and $Y_{(1)} < \cdots < Y_{(n)}$ respectively. Let $\mathbb{F}_n$ and $\mathbb{G}_n$ denote respectively the empirical distribution functions of the X and Y data, that is,

$$\mathbb{F}_n(x) = n^{-1} \sum_{1 \leq i \leq n} \mathbb{I}(X_{(i)} \leq x)$$

with an analogous expression for $\mathbb{G}_n$. Here and elsewhere in the paper, $\mathbb{I}(\cdot)$ denotes the indicator function. Observe that

$$\mathbb{F}_n(X_{(j)}) = j/n.$$

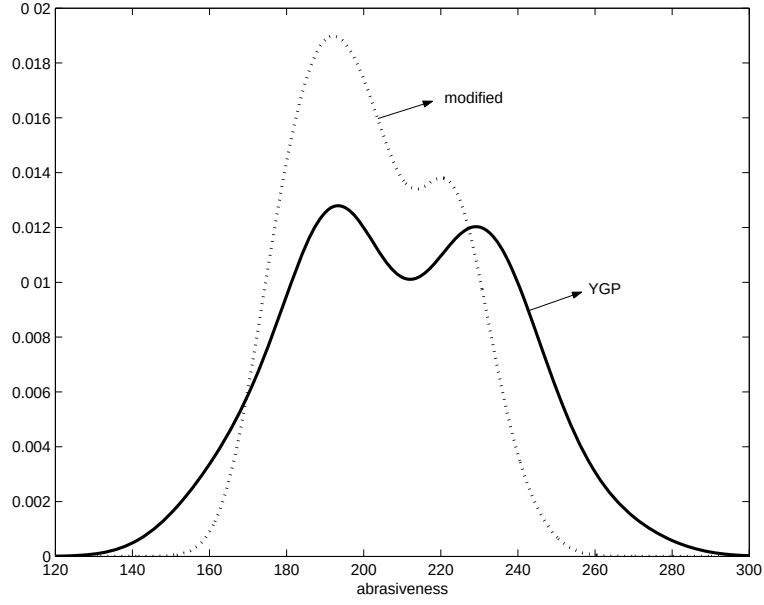


Figure 1. Kernel density estimates of marginal distributions for the coal data

The natural empirical analogue of $\Delta(x)$ in (4) is

$$\Delta_n(x) = \mathbb{G}_n^{-1} \mathbb{F}_n(x) - x$$

where $\mathbb{G}_n^{-1}$ denotes any inverse of $\mathbb{G}_n$ with the property that

$$\mathbb{G}_n^{-1}\left(\frac{j}{n}\right) = Y_{(j)}, \quad j = 1, \dots, n.$$

Then

$$\Delta_n(X_{(j)}) = Y_{(j)} - X_{(j)}, \quad j = 1, \dots, n,$$

and a scatterplot of the pairs $(X_{(j)}, Y_{(j)} - X_{(j)})$, $j = 1, \dots, n$ serves as a nonparametric estimate of Δ . Figure 2 shows this scatterplot for the abrasiveness data discussed above.

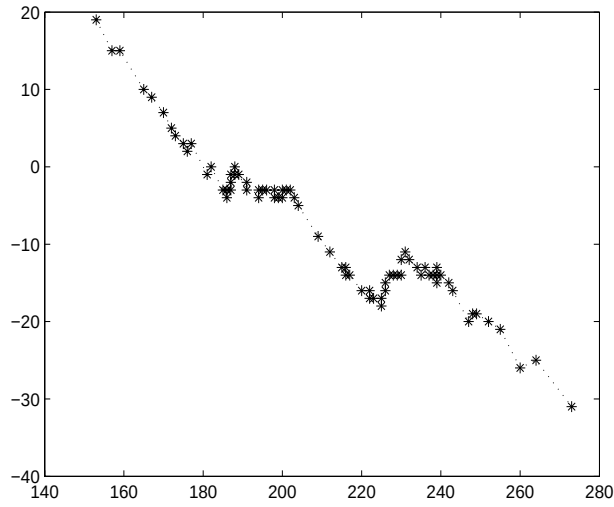


Figure 2. Nonparametric estimate of Δ

We will show in Section 5 that it is reasonable to model Δ for these data as a straight line and we will also estimate the slope and intercept of the line.

3. Tests for linearity

An informal test of the hypothesis $H_0 : q$ is linear, would be to plot the pairs of order statistics of the standardized X and Y data, namely $\tilde{Y}_{(j)} = (Y_{(j)} - \bar{Y})/S_Y$ and $\tilde{X}_{(j)} = (X_{(j)} - \bar{X})/S_X$, $j = 1, \dots, n$, and to check visually to what extent the resulting scatter clusters around the 45° line. This follows directly from the relation $(Y - \mu_Y)/\sigma_Y =_d (X - \mu_X)/\sigma_X$.

Here, $\bar{X}$, $\bar{Y}$, S_X and S_Y denote the respective sample means and standard deviations while μ_X , μ_Y , σ_X and σ_Y denote the corresponding population counterparts. This argument suggests

$$K = \sum_{j=1}^n (\tilde{X}_{(j)} - \tilde{Y}_{(j)})^2 \quad (5)$$

as a formal test statistic. A second test suggests itself upon noticing that

$$\frac{dq(x)}{dx} = f(x) / g(q(x))$$

is constant ($= \gamma_1$) when q is given by (3). Thus, an appropriate test statistic would be

$$L = n^{-1} \sum_{j=1}^n (r_j - \bar{r})^2 \quad (6)$$

where

$$r_j = n^{2/5} \log \frac{\tilde{f}(X_{(j)})}{\tilde{g}(Y_{(j)})}$$

and $\bar{r} = n^{-1} \sum r_k$. Here, $\tilde{f}$ and $\tilde{g}$ are appropriate estimates of the densities f and g and the norming constants n^{-1} and $n^{2/5}$ are suggested by asymptotic theory.

The large-sample distributions of both K and L depend in a complicated manner on the unknown joint distribution function H of X and Y . (These large-sample distributions are not discussed here. Details can be found in Potgieter (2006).) Thus, there is little hope of obtaining useful critical values via large-sample approximations. Instead, we use a permutation method to obtain p -values. To this end, notice that the data can be expressed as

$$(X_i, Y_i) = (\tilde{X}_i S_X + \bar{X}, \tilde{Y}_i S_Y + \bar{Y}), \quad i = 1, \dots, n.$$

We fix $\bar{X}$, $\bar{Y}$, S_X and S_Y and base our inferences on the distribution that results from permuting randomly the members of each of the standardized pairs $(\tilde{X}_i, \tilde{Y}_i)$ giving, say, $(\tilde{X}_i^*, \tilde{Y}_i^*)$. Thus, conditional upon the observed data, we have $(\tilde{X}_i^*, \tilde{Y}_i^*) = (\tilde{X}_i, \tilde{Y}_i)$ or $(\tilde{X}_i^*, \tilde{Y}_i^*) = (\tilde{Y}_i, \tilde{X}_i)$ with probability $1/2$ each for $i = 1, \dots, n$. This procedure produces a corresponding data set

$$(X_i^*, Y_i^*) = (\tilde{X}_i^* s_X + \bar{X}, \tilde{Y}_i^* s_Y + \bar{Y}), \quad i = 1, \dots, n.$$

The test statistic, say K^* , is calculated on the (X^*, Y^*) data set. The random permuting of the members of each pair is repeated a large number, N , of times, yielding test statistic values $K_1^*, \dots, K_N^*$. The estimated p -value is the proportion of these K^* values that exceeds the observed value, K_{obs} , of the test statistic. Exactly the same procedure is followed in respect of the test statistic L .

The observed values of K and L for the application data in Section 5 are $K_{obs} = 2.232$ and $L_{obs} = 4.445$. The corresponding estimated p -values, obtained in the manner described above, are 0.092 and 0.099 respectively. These p -values suggest that the assumption of a linear QMF is quite plausible.

The relative powers of the two test statistics were evaluated by Monte Carlo simulation against two alternatives. The first alternative is

$$\begin{aligned} q_1(x) = & \mu_X + \sigma_X (s(x) \mathbb{I}\{|s(x)| \leq 0.85\} \\ & + (as^2(x) + (1 - 1.7a)s(x) + 0.7225a) \mathbb{I}\{s(x) > 0.85\} \\ & + (-as^2(x) + (1 - 1.7a)s(x) - 0.7225a) \mathbb{I}\{s(x) < -0.85\}) \end{aligned}$$

where

$$s(x) = \frac{x - \mu_X}{\sigma_X}$$

and where a is a positive constant. Under this alternative, the QMF is the identity function on the interval $[\mu_X - 0.85\sigma_X; \mu_X + 0.85\sigma_X]$ (using

a normal approximation, 60% of the observed data will lie in this interval), with quadratic tails outside the interval. Under the alternative hypothesis, the distribution of the Y data will have heavier tails than that of the X data. The larger the value of a , the heavier the tails. Figure 3 shows q_1 for three different values of a .

The second alternative is

$$q_2(x) = \mu_X + \sigma_X \left(s(x) + \frac{b}{\pi} \left\{ \cos \left(\frac{\pi}{b} x \right) + 1 \right\} \mathbb{I} \{ |s(x)| \leq b \} \right)$$

where b is a positive constant. Under this alternative, q_2 deviates from the identity function only on the interval $[\mu_X - \sigma_X b, \mu_X + \sigma_X b]$. The deviation takes the form of a “cosine bump”, as illustrated in Figure 4 for three different values of b .

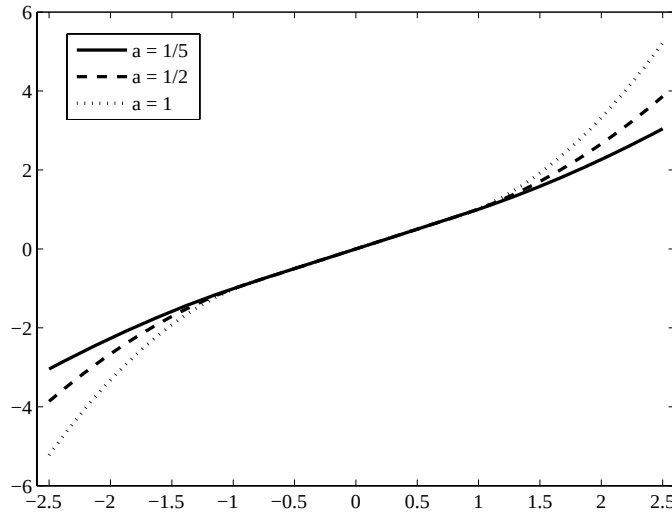


Figure 3. QMF alternative 1

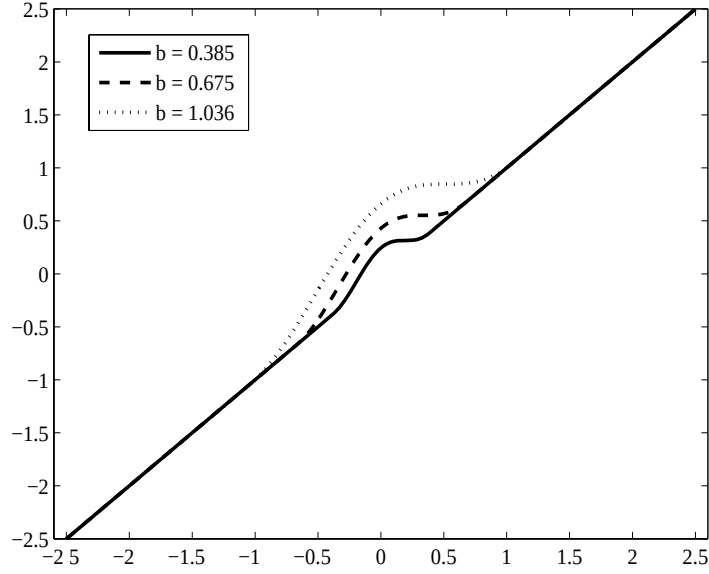


Figure 4. QMF alternative 2

For these simulations, samples were drawn from two distributions. The first distribution is a standard bivariate normal with correlation coefficient 0.4. The second distribution, referred to below as “the coal distribution”, is constructed from the application data in Section 2 in the following manner. Choose randomly a pair, denoted by (V, W) , from the set of 196 pairs

$$S = \{(X_1, Y_1), \dots, (X_{98}, Y_{98}), (Y_1, X_1), \dots, (Y_{98}, X_{98})\}.$$

This random selection defines a discrete uniform distribution on S . To make the distribution continuous, add to (V, W) a bivariate normal pair (ε, δ) with correlation coefficient 0.7 (equal to the sample correlation in the application

data) and standard deviation $h = s \times 196^{-1/5}$, where s is the standard deviation of the values $X_1, \dots, X_{98}, Y_1, \dots, Y_{98}$. The coal distribution is defined to be the distribution of the random vector $(V + \epsilon, W + \delta)$. This distribution is symmetric and continuous with identical marginals.

In the simulations to assess power, samples $(X_1, Y_1), \dots, (X_n, Y_n)$ are drawn from these distributions and each Y -value is replaced by $q_i(Y)$ ($i = 1$ or 2 depending on which alternative is being considered). The resulting sample $(X_1, q_i(Y_1)), \dots, (X_n, q_i(Y_n))$ is from a distribution with QMF equal to q_i . The power was estimated as follows. One thousand data sets were generated. In each instance, one thousand random permutations were used to obtain a p -value. Denote the 1 000 p -values obtained thus by $p_1, \dots, p_{1000}$. The estimated power is

$$\frac{1}{1000} \times (\text{number of } p_i \leq \alpha)$$

where α is the specified level of significance. We show the simulation results for $\alpha = 0.05$ in Tables 1 through 4. It is seen that K has consistently higher power than L , the difference being marginal under the alternative q_1 and more substantial under the alternative q_2 .

Table 1. Monte Carlo estimates of power against alternative q_1 : normal distribution

n	Test statistic	
	L	K
	$a = 1/5$	
50	0.084	0.086
100	0.116	0.115
250	0.190	0.193
	$a = 1/2$	
50	0.158	0.175
100	0.293	0.305
250	0.636	0.677
	$a = 1$	
50	0.299	0.403
100	0.605	0.654
250	0.949	0.973

Table 2. Monte Carlo estimates of power against alternative q_1 : coal distribution

n	Test statistic	
	L	K
	$a = 1/5$	
50	0.093	0.105
100	0.135	0.129
250	0.203	0.198
	$a = 1/2$	
50	0.189	0.209
100	0.325	0.351
250	0.718	0.740
	$a = 1$	
50	0.363	0.430
100	0.654	0.687
250	0.983	0.988

Table 3. Monte Carlo estimates of power against alternative q_2 : normal distribution

n	Test statistic	
	L	K
	$b = 0.385$	
50	0.045	0.046
100	0.062	0.076
250	0.093	0.095
	$b = 0.675$	
50	0.113	0.152
100	0.168	0.230
250	0.435	0.577
	$b = 1.036$	
50	0.279	0.372
100	0.562	0.729
250	0.966	0.987

Table 4. Monte Carlo estimates of power against alternative q_2 : coal distribution

n	Test statistic	
	L	K
	$b = 0.385$	
50	0.057	0.071
100	0.054	0.067
250	0.068	0.085
	$b = 0.675$	
50	0.085	0.134
100	0.109	0.231
250	0.219	0.467
	$b = 1.036$	
50	0.259	0.393
100	0.441	0.679
250	0.891	0.988

4. Point Estimation and confidence bands

4.1 Method of Moments

We assume throughout that moments up to order four are finite and, to avoid technical complications, that the marginal distribution functions are continuous and strictly increasing. Since no specific functional form or parameterisation of the marginal CDFs is assumed, we use the method of moments to estimate the parameters γ_0 and γ_1 in the linear QMF model (3). We also assume as a matter of convenience, that γ_1 in (3) is positive. Since $Y \stackrel{\mathcal{D}}{=} \gamma_0 + \gamma_1 X$ we see that

$$\mu_Y = \gamma_0 + \gamma_1 \mu_X \text{ and } \sigma_Y = \gamma_1 \sigma_X.$$

It follows that the moment estimator of $\Delta(x)$ from (4) is

$$\hat{\Delta}(x) = \bar{Y} - \bar{X} + (S_Y - S_X)(x - \bar{X})/S_X. \quad (7)$$

We show in Appendix 1 that $n^{1/2}(\hat{\Delta}(x) - \Delta(x))$ is asymptotically normally distributed with mean zero and variance

$$\tau^2(x) = \sigma_Y^2 \left\{ \sigma_a^2 + 2\sigma_{ab}(x - \mu_X)/\sigma_X + \sigma_b^2(x - \mu_X)^2/\sigma_X^2 \right\} / n \quad (8)$$

where

$$a = (\bar{Y} - \mu_Y)/\sigma_Y - (\bar{X} - \mu_X)/\sigma_X, \quad b = S_Y/\sigma_Y - S_X/\sigma_X$$

and

$$\begin{aligned} \sigma_a^2 &= 2(1 - \rho_{XY}) \\ \sigma_b^2 &= 1 - \rho_{XY}^2 + \{ \kappa^{4,0}/\sigma_X^4 + \kappa^{0,4}/\sigma_Y^4 - 2\kappa^{2,2}/(\sigma_X^2\sigma_Y^2) \} / 4 \\ \sigma_{ab} &= \{ \kappa^{3,0}/\sigma_X^3 + \kappa^{0,3}/\sigma_Y^3 - \kappa^{2,1}/(\sigma_X^2\sigma_Y) - \kappa^{1,2}/(\sigma_X\sigma_Y^2) \} / 2. \end{aligned} \quad (9)$$

Here $\kappa^{i,j}$ denotes the cumulant of order (i, j) in the joint distribution of X and Y – see Davison and Hinkley (1997, page 552).

Denote by $\hat{\tau}^2(x)$ the estimator of $\tau^2(x)$ that results upon substituting into (8) the corresponding sample moments and cumulants. Then, if a constant c_α satisfying the relation

$$Pr \left\{ \sup_x |\hat{\Delta}(x) - \Delta(x)| / \hat{\tau}(x) \leq c_\alpha \right\} = 1 - \alpha$$

can be found, a $100(1 - \alpha)\%$ simultaneous confidence band for Δ is determined by the inequalities

$$\hat{\Delta}(x) - c_\alpha \hat{\tau}(x) \leq \Delta(x) \leq \hat{\Delta}(x) + c_\alpha \hat{\tau}(x) \quad (10)$$

holding for all $-\infty < x < \infty$. We now show that $c_\alpha \approx (-2 \log \alpha)^{1/2}$ for large n . Let

$$\tilde{a} = n^{1/2}a/\hat{\sigma}_a, \quad \tilde{b} = n^{1/2}b/\hat{\sigma}_b, \quad \hat{\rho}_{ab} = \hat{\sigma}_{ab}/(\hat{\sigma}_a\hat{\sigma}_b)$$

and

$$\tilde{C}^2 = [\tilde{a} \ \tilde{b}] \widehat{\Sigma}^{-1} [\tilde{a} \ \tilde{b}]' \quad (11)$$

where

$$\widehat{\Sigma} = \begin{bmatrix} 1 & \hat{\rho}_{ab} \\ \hat{\rho}_{ab} & 1 \end{bmatrix}$$

is the estimated covariance matrix of the vector $[\tilde{a} \ \tilde{b}]'$. Some algebraic manipulation then shows that

$$\sup_x |\hat{\Delta}(x) - \Delta(x)| / \hat{\tau}(x) \leq c_\alpha$$

if and only if

$$\tilde{C}^2 \sigma_Y^2 / S_Y^2 \leq c_\alpha^2.$$

From (11) and the joint asymptotic bivariate normality of $[\tilde{a} \ \tilde{b}]$ it follows that $\tilde{C}^2$ has asymptotically a chi-squared distribution with 2 degrees of freedom.

Furthermore, S_Y^2/σ_Y^2 converges in probability to 1. Thus

$$\alpha = P \left\{ \tilde{C}^2 \sigma_Y^2 / S_Y^2 < c_\alpha^2 \right\} \approx 1 - \exp(-c_\alpha^2/2)$$

for large n . Hence, $c_\alpha \approx (-2 \log \alpha)^{1/2}$ for large n .

The quality of this approximation to c_α cannot be expected to be satisfactory in general because sample estimates of the third and fourth order cumulants which figure in the calculation of $\hat{\tau}(x)$ are known to be highly variable. Thus, the rate of convergence to normality would typically be rather slow. An empirical estimate, c_α^* , of c_α can be made in any specific instance by estimating the distribution of $\tilde{C}^2\sigma_Y^2/S_Y^2$ from bootstrap samples $(X_{1,j}^*, Y_{1,j}^*), \dots, (X_{n,j}^*, Y_{n,j}^*)$, $j = 1, \dots, B$, drawn with replacement from the data $(X_1, Y_1), \dots, (X_n, Y_n)$. Because $\tilde{C}^2\sigma_Y^2/S_Y^2$ is a pivot, a bootstrap approximation to c_α will generally be more accurate in samples of moderately large size than will the large-sample approximation – see Davison and Hinkley (1997, page 29) and the simulation results in Section 5 below.

Remark The following type of measurement error model often arises in practice. Let t denote the “true” YGP value of a batch of coal. If all coal samples were subjected to a YGP analysis, the respective determinations X and X' of t would have the structure

$$X = t + \epsilon, \quad X' = t + \epsilon', \quad (12)$$

where the *measurement errors* ϵ and ϵ' are independently and identically distributed, independently of t . Typically, ϵ and ϵ' would be normally distributed but t might not be so distributed – see Figure 1. If one sample from each pair is analyzed instead with the modified YGP method and if the effect of this “treatment” is assumed to be a location-scale change on X' , then the observed value Y would have the structure $Y = \gamma_0 + \gamma_1 X'$. Then $q(x) = \gamma_0 + \gamma_1 x$ and we have a linear QMF model.

Now, assume normally distributed measurement errors ϵ and ϵ' in (12) and recall that all their cumulants of order greater than 2 vanish. Straightforward calculation based upon (12) then shows that all terms in (9) involving cumulants of order 3 and 4 *cancel*, notwithstanding the fact that no

assumption of normality in the distribution of t has been made. Then σ_{ab} in (8) is zero and $\sigma_b^2 = 1 - \rho^2$. The estimated variance of $\hat{\Delta}(x)$ in (7) becomes

$$\hat{\tau}^2(x) = S_Y^2(1 - r) \left\{ 2 + (1 + r_{XY}) (x - \bar{X})^2 / S_X^2 \right\} / n$$

which, in the absence of the positive covariance term $\hat{\sigma}_{ab}$, results in a much narrower confidence band for Δ than that provided by (8) and (10).

4.2 Method of Quantiles

In view of the large variability of higher order moments mentioned in the preceding section, it may be useful to construct a confidence band for $\Delta(x)$ that is robust against outlying values. This suggests a quantile-based estimator of $\Delta(x)$, namely

$$\hat{\Delta}(x) = \tilde{Y} - \tilde{X} + (\tilde{S}_Y - \tilde{S}_X)(x - \tilde{X}) / \tilde{S}_X \quad (13)$$

where $\tilde{X}$, $\tilde{Y}$, $\tilde{S}_X$ and $\tilde{S}_Y$ denote the respective sample medians and interquartile ranges of the X and Y data. The population medians and interquartile ranges are denoted below by m_X , m_Y , i_X and i_Y respectively. The latter quantities exist under all circumstances and the assumption of finite moments to order four (made in the preceding section) is thus obviated.

One can show (see Potgieter, 2006) that

$$n^{1/2} \left(\hat{\Delta}(x) - \Delta(x) \right)$$

is asymptotically normally distributed with variance function

$$\tau^2(x) = i_Y^2 \left\{ \sigma_a^2 + 2\sigma_{ab}(x - m_X) / i_X + \sigma_b^2 (x - m_X)^2 / i_X^2 \right\} / n. \quad (14)$$

The expressions for σ_a^2 , σ_b^2 and σ_{ab} are given in Appendix 2. The $100(1 - \alpha)\%$ confidence band for Δ has exactly the same form as that in (10).

Again, the corresponding asymptotic estimate of c_α in (10) would typically not be appropriate for use in small samples and recourse may be

had to the bootstrap. However, since the variance terms involve the unknown density functions f and g , a smoothed bootstrap would have to be used – see Davison and Hinkley (1997, pp. 79–81 and Problem 8, page 127).

4.3 Comparison of estimators

We did a small Monte Carlo study to compare the estimators at various sample sizes for four underlying bivariate distributions. The following table gives a summary of the biases and standard errors of the estimators. All distributions were standardised to have mean 0 and variance 1. Distribution 1 is a bivariate normal with correlation 0.4. Distribution 2 is the coal distribution – short-tailed and bimodal. Distribution 3 is a bivariate distribution with Normal Inverse Gaussian marginals – skewed to the right. Distribution 4 is a bivariate distribution with marginals that have a t -distribution with 5 degrees of freedom – symmetric, but heavy-tailed. In the latter two instances, a Gaussian copula was assumed.

The estimators based on moments are superior in terms of standard error for distributions 1 through 3 – light tailed and/or skew. This is not too unexpected and doubtless the result of the “central limit theorem effect”. In the fourth distribution (heavy-tailed), the estimators based on quantiles are superior in terms of standard error at the sample sizes considered here. This is also not too unexpected because the “central limit theorem effect” manifests itself only at comparatively large sample sizes when the underlying distributions are heavy tailed.

n	Population 1 (Method of Moments)				Population 1 (Method of Quantiles)			
	γ_0		γ_1		γ_0		γ_1	
	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$
50	-0.0122	1.1169	0.0571	0.9410	0.0205	1.5680	0.2563	1.7408
100	-0.0175	1.1070	0.0455	0.9311	0.0035	1.5452	0.2012	1.6669
250	-0.0270	1.0913	0.0221	0.9059	-0.0163	1.5141	0.1190	1.6214
	Population 2 (Method of Moments)				Population 2 (Method of Quantiles)			
	γ_0		γ_1		γ_0		γ_1	
	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$
50	0.0128	0.8974	-0.3063	0.8183	-0.2264	1.4877	0.0002	1.3374
100	0.0099	0.8872	-0.4814	0.8020	-0.4302	1.4536	-0.1160	1.2965
250	0.0133	0.8764	-0.8074	0.7943	-0.7408	1.4271	-0.2720	1.2412
	Population 3 (Method of Moments)				Population 3 (Method of Quantiles)			
	γ_0		γ_1		γ_0		γ_1	
	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$
50	0.0835	1.1825	0.1732	1.5863	0.0905	1.4619	0.2253	1.8681
100	0.0639	1.2418	0.1362	1.6439	0.0762	1.5069	0.1776	1.8332
250	0.0700	1.4143	0.0953	1.7344	0.0657	1.6693	0.1042	1.9085
	Population 4 (Method of Moments)				Population 4 (Method of Quantiles)			
	γ_0		γ_1		γ_0		γ_1	
	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$	$\sqrt{n}Bias$	$\sqrt{n}SE$
50	0.0226	1.3244	0.2743	1.9703	0.0176	1.3633	0.1816	1.7808
100	0.0151	1.4016	0.2273	2.1507	0.0128	1.4018	0.1140	1.7634
250	0.0198	1.6291	0.1780	2.4923	0.0267	1.5323	0.0742	1.7806

5. Data analysis

Figure 5 shows the simultaneous 95% bootstrap confidence band (10) found from the abrasiveness data. The non-parametric band constructed by Lombard (2005, Section 4) is also shown for comparison. (The non-parametric estimate of Δ was shown in Figure 2). The bands are plotted against the standardized value $v = (x - \bar{X})/S_X$. The band from the linear QMF model is uniformly substantially narrower than the non-parametric band and, in contrast to the

latter, extends over the full range of the data. Since the band does not contain the origin and also intersects the v -axis, we conclude at the 5% level of significance that the modified YGP method caused a decrease in *both* the mean and the variability of the abrasiveness value. The band from the linear QMF model is based upon an estimate $c_{0.05}^* = 2.53$ of $c_{0.05}$ found from $B = 1,000$ bootstrap samples, each of size 98, drawn with replacement from the 98 pairs of abrasiveness values. Asymptotic theory gives the smaller critical value

$$c_{0.05} = (-2 \log 0.05)^{1/2} = 2.45.$$

Some simulations were run to gauge the appropriateness of asymptotic and bootstrap critical values in this particular application. Ten thousand data sets were sampled from the estimated copula for these data that were constructed by Lombard (2005, Section 4). For each of these data sets c_α was estimated from 1,000 bootstrap samples. Table 5 shows Monte Carlo the estimate $\hat{p}(\alpha)$ of $P(T \leq (-2 \log \alpha)^{1/2})$ and the estimate $\hat{p}^*(\alpha)$ of $P(T \leq c_\alpha^*)$ for $n = 98$ and $n = 25$. While the bootstrap results are quite acceptable, use of the asymptotic critical value produces consistently an estimated coverage probability well below the nominal value.

Table 5. Comparison of coverage probabilities from asymptotic theory and the bootstrap

	$n = 25$			$n = 98$		
$1 - \alpha$	0.90	0.95	0.99	0.90	0.95	0.99
$\hat{p}(\alpha)$	0.773	0.834	0.910	0.824	0.895	0.962
$\hat{p}^*(\alpha)$	0.911	0.964	0.996	0.916	0.961	0.994

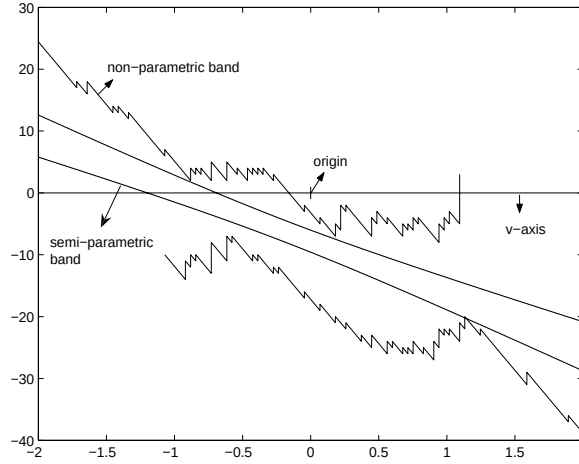


Figure 5. Semi-parametric and non-parametric simultaneous confidence bands for Δ

6. Concluding remarks

1. A natural invariance requirement is that our inferences should remain unchanged if the data are relabelled $(X, Y) \rightarrow (Y, X)$. Put another way, the upper (lower) limit of the band for $q(x) - x$ should equal the lower (upper) limit of the band for $q^{-1}(x) - x$. That this is not generally true for the construction given in Section 4 is clear upon inspection of (11). For while $\tilde{C}^2$ there remains unchanged under such a relabelling, the factor σ_Y^2/S_Y^2 must be replaced by σ_X^2/S_X^2 . Of course, the difference between the latter two quantities is negligible in large samples because both are then close to 1. (In the example considered in Section 5, the difference is indeed

inconsequential). In smaller samples, however, the discrepancy may be more material. Nonetheless, the defect can be remedied in the following way.

Denote by B^* the reflection of a set B of points in the plane around the x (horizontal) -axis. Let B_X be the set of points in the plane delineated by (10). Then, if the data are relabelled $(X, Y) \rightarrow (Y, X)$, the invariance requirement is that $B_Y^* = B_X$. Now, define the constant c_α figuring in (10) by

$$Pr \left\{ \tilde{C}^2 \times \max\{\sigma_X^2/S_X^2, \sigma_Y^2/S_Y^2\} \leq c_\alpha \right\} = 1 - \alpha,$$

where $\tilde{C}^2$ is given in (11), and take $B_X \cap B_Y^*$ as the confidence band.

2. Denote by $\beta_{Y|X}$ and by $\beta_{X|Y}$ respectively the slopes of the linear regression lines of Y upon X and of X upon Y and write $q(x) = \gamma_0 + \gamma_1 x$. Both regression lines, as well as q , pass through the point (μ_X, μ_Y) and it is easily seen from the relations

$$\beta_{Y|X} = \rho\sigma_Y/\sigma_X \text{ and } \beta_{X|Y} = \rho\sigma_X/\sigma_Y$$

that

$$\beta_{Y|X} < \gamma_1 < 1/\beta_{X|Y}$$

when ρ , the correlation coefficient between X and Y , is positive. Thus, we see that q lies between the two regression lines.

3. The methodology proposed in this paper applies to any family of distributions that can be brought into location-scale form by a suitable one-to-one transformation. For instance, a scaled power-law family $G(x) = F(\sigma x^\gamma)$, $x > 0$, can be accommodated by the simple device of taking logarithms of the observations.

4. The methodology proposed in Section 4 generalizes readily to situations involving a control and *two or more* treatments. If $k+1$ variables $(X, Y_1, \dots, Y_k)$ are involved, denote by $\tilde{C}_k$ the $\tilde{C}$ in (11) with Y replaced

by Y_k . If c_α satisfies

$$P \left\{ \max_{1 \leq j \leq k} \tilde{C}_j \sigma_{Y_j}^2 / S_{Y_j}^2 \leq c_\alpha^2 \right\} = 1 - \alpha,$$

then the k pairwise bands (each computed with X as control) together give a joint coverage probability of $1 - \alpha$. A bootstrap estimate of c_α can be found using an obvious generalisation of the method described in Section 4.

7. Summary

In this paper the estimation of treatment effects from matched pair data has been considered. When the copula is symmetric, the treatment effect is expressible *solely* in terms of the relationship between the marginal distributions. We argued that expression of the treatment effect in terms of the marginal distributions is often justified, despite the fact that such expression does not involve the dependence structure within pairs. The *statistical properties* of the estimators of treatment effect do, however, involve the dependence structure.

Situations in which the treatment effect consists of a location and/or scale change have been singled out for attention in this paper. In this case, a quantile comparison function proposed by Lehmann (1974) has a linear form. A graphical device, supplemented by a permutation test, has been proposed to identify such situations. A simultaneous confidence band for the quantile comparison function has been constructed, which allows for a straightforward graphical assessment of the nature of the treatment effect. The implementation of the method in situations involving two or more treatments has also been indicated.

Situations in which the response to a treatment is multivariate have not been considered in the present paper. Flury, Nel and Pienaar (1995) devised a mode of analysis that is applicable in fully randomized designs, assuming

multivariate normality of the response vector. They also show that their analysis remains applicable if the normality assumption is not met, provided the parametric test procedures are replaced by appropriate permutation tests. It is an open question whether a similar generalization of the problem considered in the present paper can be made.

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Appendix 1. Asymptotic distribution of $\hat{\Delta}(x)$

Set

$$a = (\bar{Y} - \mu_Y)/\sigma_Y - (\bar{X} - \mu_X)/\sigma_X, \quad b = S_Y/\sigma_Y - S_X/\sigma_X.$$

Then

$$\hat{\Delta}(x) - \Delta(x) = \sigma_Y (a + b(x - \bar{X})/S_X).$$

It is easy to show, using the central limit theorem on $\bar{X}$ and the consistency of S_X as an estimator of σ_X , that

$$b(x - \bar{X})/S_X = b(x - \mu_X)/\sigma_X + o_P(n^{-1/2}),$$

whence $n^{1/2}(\hat{\Delta}(x) - \Delta(x))$ and $n^{1/2}\sigma_Y (a + b(x - \mu_X)/\sigma_X)$ have the same asymptotic distribution. Thus, we restrict attention to the latter quantity.

Set

$$x_i = (X_i - \mu_X)/\sigma_X, \quad y_i = (Y_i - \mu_Y)/\sigma_Y.$$

Clearly, in the following we may assume without any loss of generality that

$$\mu_X = \mu_Y = 0; \quad \sigma_X = \sigma_Y = 1.$$

Straightforward application of the central limit theorem and the delta method – see, e.g. Ferguson (1996, Theorem 8, page 52) – shows that the vector $[\bar{x} \ \bar{y} \ S_x \ S_y]'$ is asymptotically normally distributed with mean $[0 \ 0 \ 1 \ 1]'$ and covariance matrix $M/n^{1/2} = \{m_{ij}/n^{1/2}\}$ where

$$\begin{aligned} m_{11} &= 1, \quad m_{12} = \rho, \quad m_{1,3} = \kappa_{x,y}^{3,0}, \quad m_{14} = \kappa_{x,y}^{1,2}, \\ m_{22} &= 1, \quad m_{23} = \kappa_{x,y}^{2,1}, \quad m_{2,4} = \kappa_{x,y}^{0,3}, \\ m_{33} &= \kappa_{x,y}^{4,0}/4 + 1/2, \quad m_{3,4} = \kappa_{x,y}^{2,2}/4 + \rho^2/2, \\ m_{44} &= \kappa_{x,y}^{0,4}/4 + 1/2 \end{aligned} \tag{15}$$

and where $\kappa_{x,y}^{i,j}$ denotes the cumulant of order (i,j) in the joint distribution of (x,y) . The map

$$[\bar{x} \ \bar{y} \ S_x \ S_y] \mapsto [\bar{y} - \bar{x} \ S_y - S_x] = [a \ b]$$

has Jacobian matrix

$$j = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

from which it follows that the asymptotic joint distribution of $n^{1/2}[a \ b]'$ is normal with mean zero and covariance matrix

$$jMj' = \begin{bmatrix} m_{11} - m_{12} + m_{22} - m_{12} & m_{24} - m_{14} + m_{13} - m_{23} \\ \cdot & m_{44} - m_{34} + m_{33} - m_{43} \end{bmatrix}.$$

Simplifying and substituting

$$\kappa_{x,y}^{i,j} = \kappa^{i,j} / (\sigma_X^i \sigma_Y^j)$$

leads to

$$jMj' = \begin{bmatrix} \sigma_a^2 & \sigma_{ab} \\ \cdot & \sigma_b^2 \end{bmatrix}$$

with σ_a^2 , σ_b^2 and σ_{ab} given in (9). Thus, the expression (8) for the asymptotic variance of $n^{1/2}(\hat{\Delta}(x) - \Delta(x))$ follows.

Appendix 2. Covariance matrix for the method of quantiles

The expressions below fare the result of an application of the functional delta method. Details can be found in Potgieter (2006).

$$\begin{aligned}
\sigma_a^2 &= \frac{1}{2i_Y^2 g^2 (G^{-1}(1/2))} + \frac{1}{2i_X^2 f^2 (F^{-1}(1/2))} \\
&\quad - 2 \frac{C(1/2, 1/2) - 1/4}{i_X i_Y f (F^{-1}(1/2)) g (G^{-1}(1/2))} \\
\sigma_b^2 &= \frac{1}{16i_Y^2} \left(\frac{3}{g^2 (G^{-1}(3/4))} + \frac{3}{g^2 (G^{-1}(1/4))} - \frac{2}{g(G^{-1}(1/4)) g(G^{-1}(3/4))} \right) \\
&\quad + \frac{1}{16i_X^2} \left(\frac{3}{f^2(F^{-1}(3/4))} + \frac{3}{f^2(F^{-1}(1/4))} - \frac{2}{f(F^{-1}(1/4)) f(F^{-1}(3/4))} \right) \\
&\quad + \frac{1}{i_X i_Y} \left(\frac{C(3/4, 3/4) - 9/16}{g(G^{-1}(3/4)) f(F^{-1}(3/4))} + \frac{C(1/4, 1/4) - 1/16}{g(G^{-1}(1/4)) f(F^{-1}(1/4))} \right. \\
&\quad \left. - \frac{C(1/4, 3/4) - 3/16}{g(G^{-1}(3/4)) f(F^{-1}(1/4))} - \frac{C(3/4, 1/4) - 3/16}{g(G^{-1}(1/4)) f(F^{-1}(3/4))} \right) \\
\sigma_{ab} &= \frac{1}{8i_Y^2} \left(\frac{1}{g(G^{-1}(1/4)) g(G^{-1}(1/2))} - \frac{1}{g(G^{-1}(1/2)) g(G^{-1}(3/4))} \right) \\
&\quad + \frac{1}{8i_X^2} \left(\frac{1}{f(F^{-1}(1/4)) f(F^{-1}(1/2))} - \frac{1}{f(F^{-1}(1/2)) f(F^{-1}(3/4))} \right) \\
&\quad - \frac{1}{i_Y i_X} \left(\frac{C(3/4, 1/2) - 3/8}{g(G^{-1}(1/2)) f(F^{-1}(3/4))} - \frac{C(1/4, 1/2) - 1/8}{g(G^{-1}(1/2)) f(F^{-1}(1/4))} \right. \\
&\quad \left. - \frac{C(1/2, 1/4) - 1/8}{g(G^{-1}(1/4)) f(F^{-1}(1/2))} + \frac{C(1/2, 3/4) - 3/8}{g(G^{-1}(3/4)) f(F^{-1}(1/2))} \right).
\end{aligned}$$